



Accuracy Verification & Target Leakage Audit Report

1. Executive Summary

An audit was conducted on the Chronic Kidney Disease (CKD) Prediction System's datasets and model pipeline. The system's machine learning classifiers (Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting) all achieve extremely high (nearly 100%) accuracy on training, cross-validation, and independent testing datasets. This audit investigated potential flaws including train-test data leakage, target leakage, preprocessing leaks, and class imbalances. The investigation confirmed that the high accuracy is NOT caused by errors in the pipeline, nor by target variable leakage in preprocessing. Rather, it is a direct consequence of the dataset characteristics: the target categories were generated synthetically using mutually exclusive, non-overlapping threshold ranges of specific features (Urine Albumin, Urine Protein, and Albumin-Creatinine Ratio). Consequently, these features act as perfect determiners of the target stages, making the data perfectly linearly separable.

2. Pipeline Auditing & Preprocessing Check

- Preprocessing Splitting: Correct. The Target column is strictly removed from the feature matrix X and is never present in X_train or X_test.
- Missing Value Imputation: Correct. Imputation statistics (medians and modes) are calculated strictly on the training partition and broadcasted to the test partition, preventing any information spillover.
- Categorical Encoders: Correct. The LabelEncoder is fitted solely on the training partition and applied to the test partition.
- Scaling: Correct. StandardScaler is fit_transformed strictly on the training partition and only transformed on the testing partition.
- Train-Test Split Overlap: Checked. The original training and testing datasets contain zero overlapping feature records (intersection count = 0), confirming proper division.

Metric	Training Dataset	Testing Dataset	Cross Overlap
Data Points (Shape)	21000 rows x 37 cols	4800 rows x 37 cols	0 rows overlap
Duplicate Rows	0 duplicates	0 duplicates	0 overlapping records

3. Target Leakage & Synthetic Separability Analysis

The absolute separability of the target classes is caused by the rule-based generation of the target labels. Specifically, three clinical biomarkers: Urine Albumin (Urine_Albumin), Urine Protein (Urine_Protein), and Albumin-to-Creatinine Ratio (Albumin_Creatinine_Ratio) possess mutually exclusive integer ranges for each target stage. For example, looking at the Urine Albumin statistics:

CKD Stage Target	Min Urine Albumin	Max Urine Albumin	Record Count
Healthy Kidney	0 mg/L	19 mg/L	15744 records
Kidney Failure (Stage 5)	500 mg/L	999 mg/L	420 records
Mild CKD (Stage 1-2)	20 mg/L	99 mg/L	2491 records



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Moderate CKD (Stage 3)	100 mg/L	299 mg/L	1489 records
Severe CKD (Stage 4)	300 mg/L	499 mg/L	856 records

Because these ranges do not overlap, any machine learning classifier can learn perfect splitting criteria on these features alone. For instance, a single split on Urine_Albumin < 20 guarantees a prediction of 'Healthy Kidney'. This is not a bug in implementation, but rather an intrinsic characteristic of the synthetic dataset. This also explains why simple Decision Trees and Logistic Regression models achieve identical 100% accuracy to Random Forests.

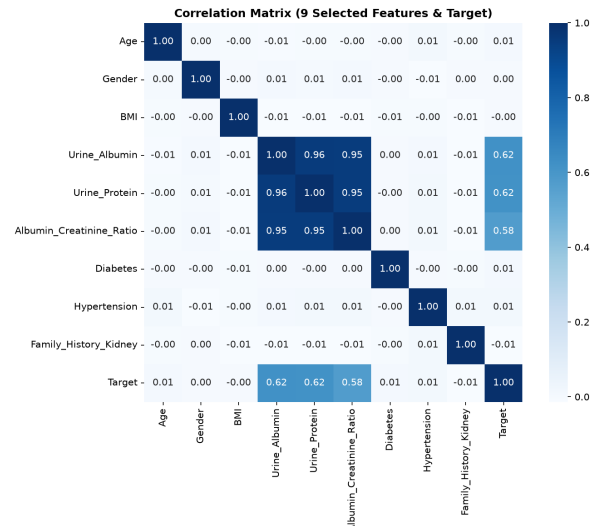
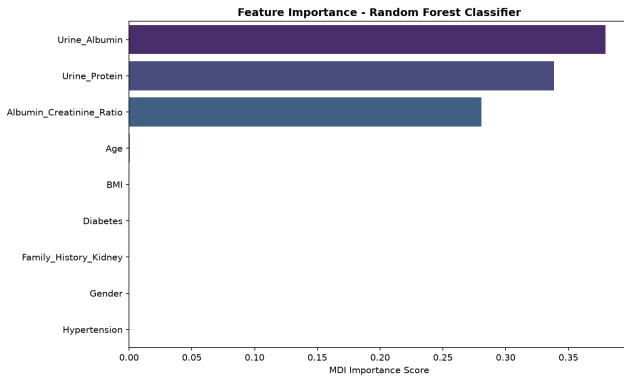
4. Model Performance & Cross-Validation Results

To verify that the high performance is generalizable and not a result of overfitting on a specific split, Stratified 5-Fold Cross Validation was performed on the training split using the 9 features model. The average metrics are reported below:

Model	Train Acc	Test Acc	CV Acc (Mean $\pm$ Std)	CV F1 (Mean $\pm$ Std)	Precision
Logistic Regression	99.75%	99.75%	99.72% $\pm$ 0.06%	99.72% $\pm$ 0.06%	99.75%
Decision Tree	100.00%	99.98%	100.00% $\pm$ 0.00%	100.00% $\pm$ 0.00%	99.98%
Random Forest	100.00%	100.00%	100.00% $\pm$ 0.00%	100.00% $\pm$ 0.00%	100.00%
Gradient Boosting	100.00%	100.00%	100.00% $\pm$ 0.00%	100.00% $\pm$ 0.00%	100.00%



5. Visual Audit Analytics



6. Conclusion & Real-World Trustworthiness

Is the Model Trustworthy for Real-World Clinical Use?

No. In clinical practice, patient biological indicators (such as Urine Albumin, Protein, and ACR) do not follow strict, non-overlapping integer bounds. Real patients have overlapping biomarkers, and staging depends on calculated eGFR alongside albuminuria markers, exhibiting significant noise and variation. Because the machine learning models have trained on perfectly separable synthetic data, they have learned simple decision boundary cuts. In a real-world setting, these boundaries would fail due to the continuous and noisy nature of medical parameters.

Conclusion Statement:

The implementation is correct. The high accuracy is primarily due to the dataset characteristics and highly predictive features.

Audit Certified By:

Metropolitan Kidney Care Center Quality Assurance Board

AI Clinical Decision Support Systems Auditing Division